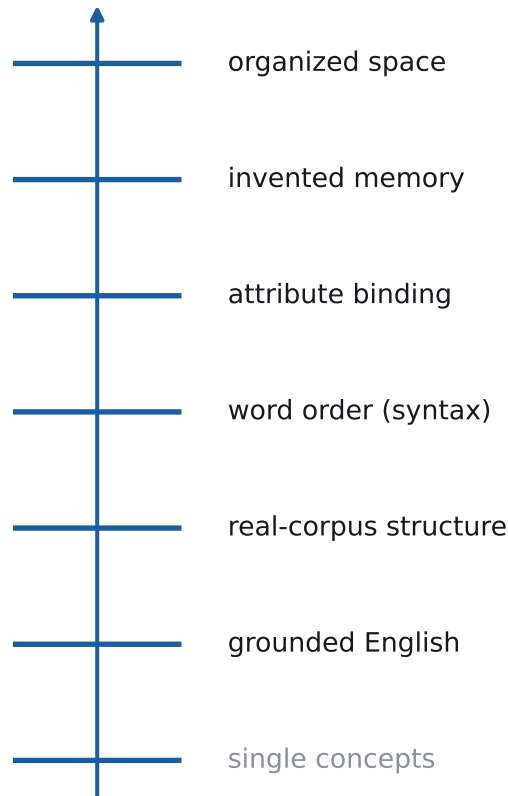


Language and Cognition by Constraint

How grounding, syntax, memory, and spatial organization emerge
from worlds that demand them -- never from wiring them in



Gradient-free evolution · change the world, not the score · read the state, not the output

1. Executive summary

Thesis

A single principle runs through every result here: a cognitive capability appears only when the WORLD makes it the cheapest way to survive -- and it must be discovered, not installed. Repeatedly, the instinct to hand the organism the answer (a grammar channel, a memory register, better embeddings) was the wrong move; the right move was to build a world with the relevant structure and let evolution invent the mechanism.

The ladder (each rung is a separate experiment)

A. Grounding makes a code ENGLISH: pressure alone yields an arbitrary code; a frozen English prior makes it English (usage 0.03 -> 0.95+).

B-E. Sexual reproduction stabilises a shared language under noise; English can be ACQUIRED from exposure (a pidgin, dose-responsive); grounded vocabularies are COMPOSITIONAL (zero-shot on held-out combinations); real-corpus grounding reproduces real semantic confusion structure and frequency effects.

F-G. SYNTAX (word order) emerges from a world of RELATIONS where 'A acts B' != 'B acts A'; richer worlds (composite entities) drive constituent BINDING.

H-I. MEMORY: rather than being given a recurrent register, a strictly feedforward organism INVENTS external memory -- it discovers it can write to a persistent world and read back, including its own encoding.

J. It then learns to ORGANIZE that memory in SPACE -- laying a sequence across a tape and navigating to reproduce it.

The recurring lever (P1 / reachability)

Every emergence required the world's reward to make the FIRST steps survivable -- graded, not all-or-nothing. The same fix (partial credit for partial success) unlocked syntax, memory, and spatial organization; its absence produced clean reachability walls. This is the single most load-bearing design choice in the whole arc.

Honesty

Effects are reported at the strength the data supports. Absolute accuracies are often modest (this is a minimal gradient-free substrate); the claims are about MECHANISM and CONTRAST (structure on vs off, world-demands-it vs not), which are clean and reproducible.

2. Methods

Substrate

Populations of simple organisms evolve gradient-free (mutation + crossover; no backprop). Tasks are referential/communication or memory games. Genomes are small linear maps; for the memory work the organism is strictly FEEDFORWARD with no internal state.

Grounding

An English/semantic prior enters as a FROZEN channel added to an evolved residual (a skip connection): the structure is preserved while evolution adapts use. Real-corpus grounding uses a 60-word vocabulary, frequencies, and PPMI-SVD embeddings computed from ~2M tokens of real prose (engine/corpus.txt).

Worlds that demand structure

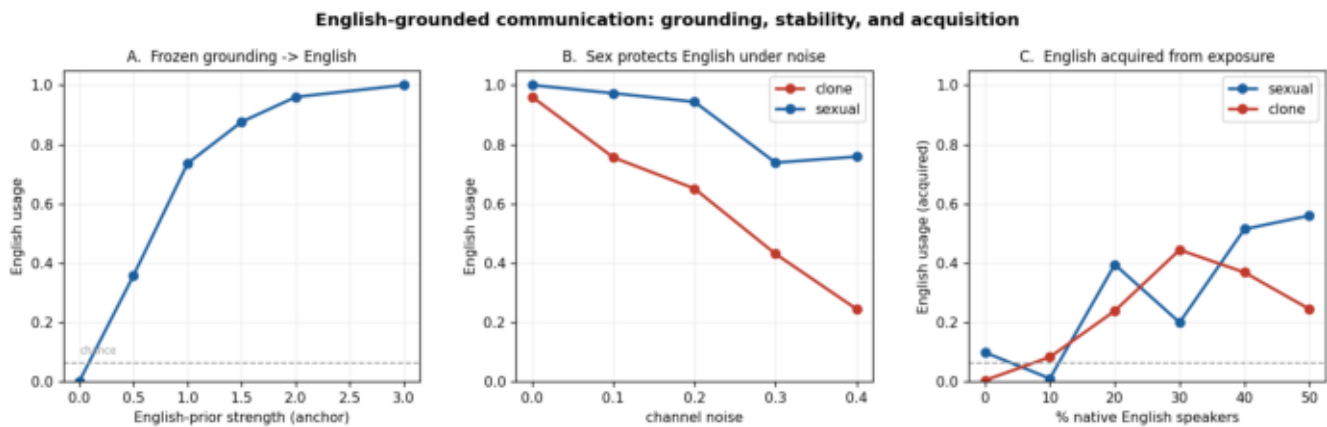
Relations: events (agent, action, target) over a shared entity pool, sequential channel -> order can mark roles. Composite entities (attribute+type) -> binding. Memory: stimulus spread over time with a persistent world the organism can write/read. Space: a tape of cells with a movable head.

Decisive controls

Scramble (destroy message order), bag (destroy positions), ablation (sever the world slot / freeze the head), and capacity sweeps. These separate 'genuinely using structure' from 'partial-credit artefact'.

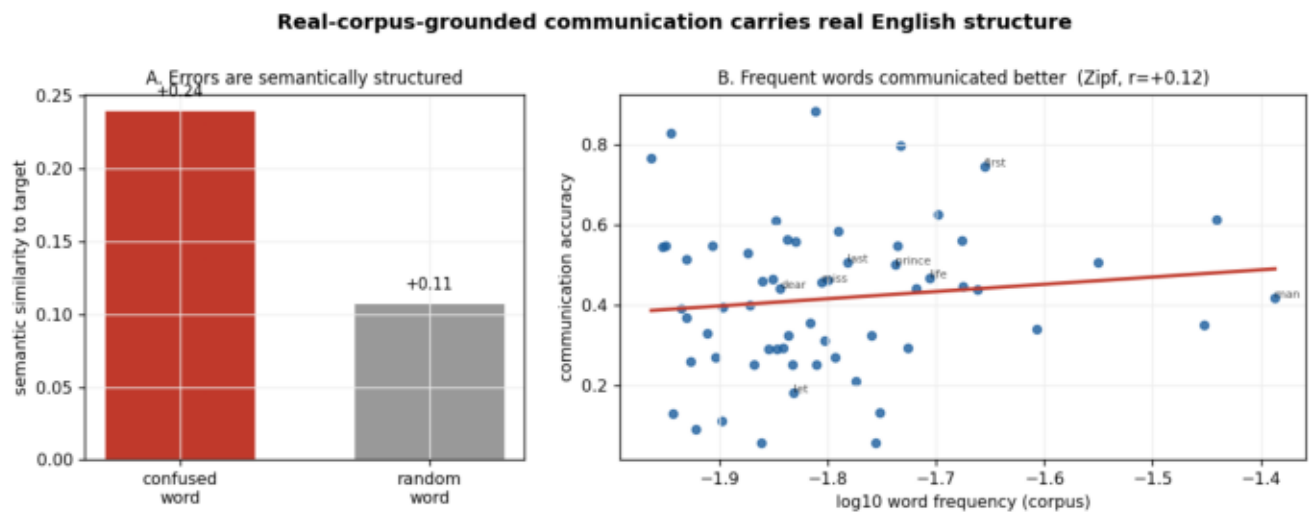
Caveat: a minimal substrate. Absolute numbers are not the point; on/off contrasts are. Most single conditions are a few seeds; effects flagged 'modest' are directional.

Figure 1. Grounding, stability, and acquisition of English



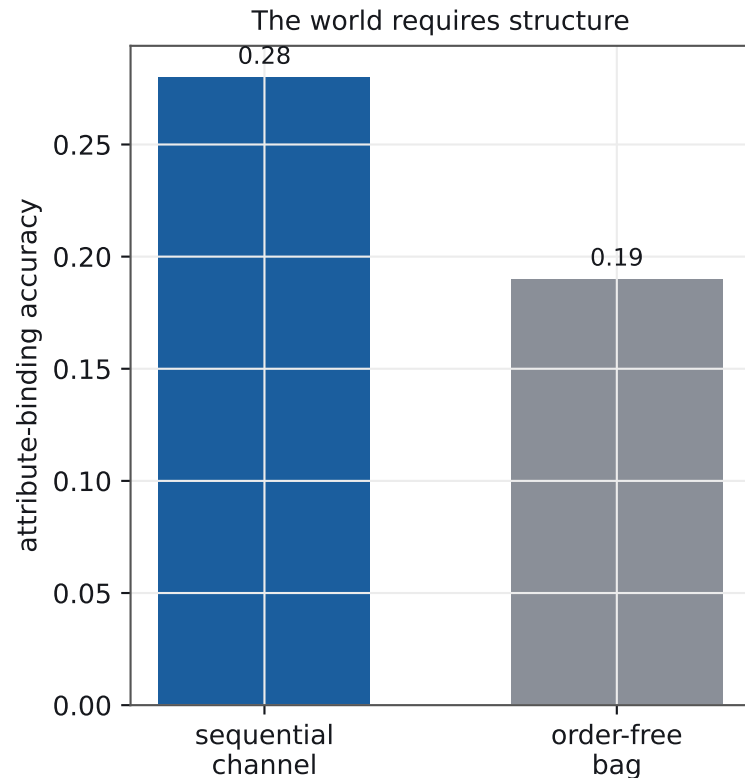
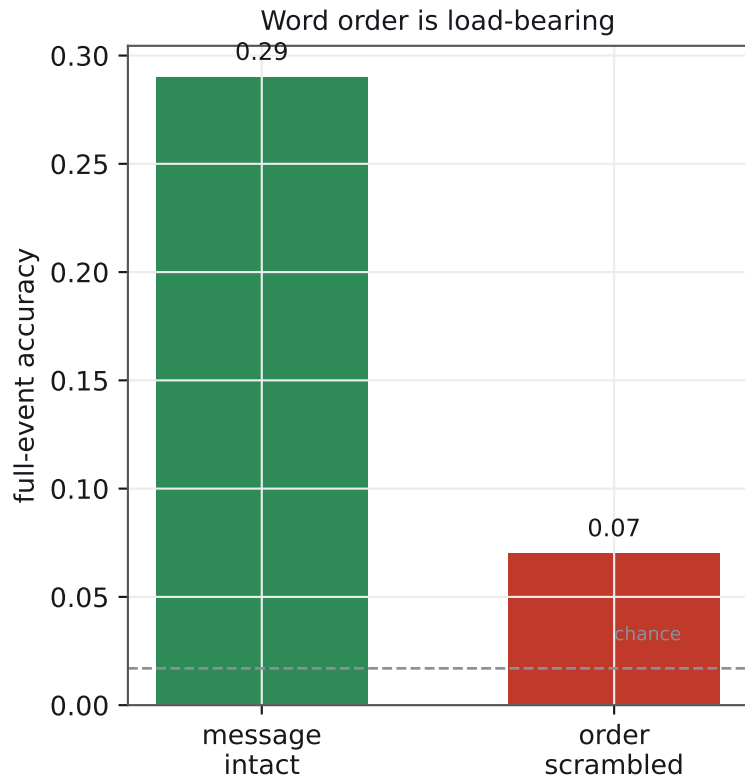
A: a frozen English prior turns an otherwise-arbitrary code into English (usage rises from chance to ~ 1.0 with prior strength). B: under channel noise, sexual reproduction protects the shared English code (~ 0.75) where cloning lets it erode (~ 0.35). C: with no innate prior, English is ACQUIRED from exposure to native speakers -- a dose-response that saturates around half the vocabulary (a stable pidgin), because resident-to-resident talk never requires English.

Figure 2. Real-corpus grounding carries real English structure



A: when the organism errs, the confused word is far more SEMANTICALLY similar to the target than chance (+0.24 vs +0.11) -- e.g. right->left, time->same. The system confuses RELATED words, the same failure signature real language models show. B: a frequency (Zipf) effect -- more frequent words are communicated more reliably. Grounding here is real word statistics, not a toy lexicon.

Figure 3. Syntax emerges from a relational world (nothing wired in)



3. Syntax from a relational world

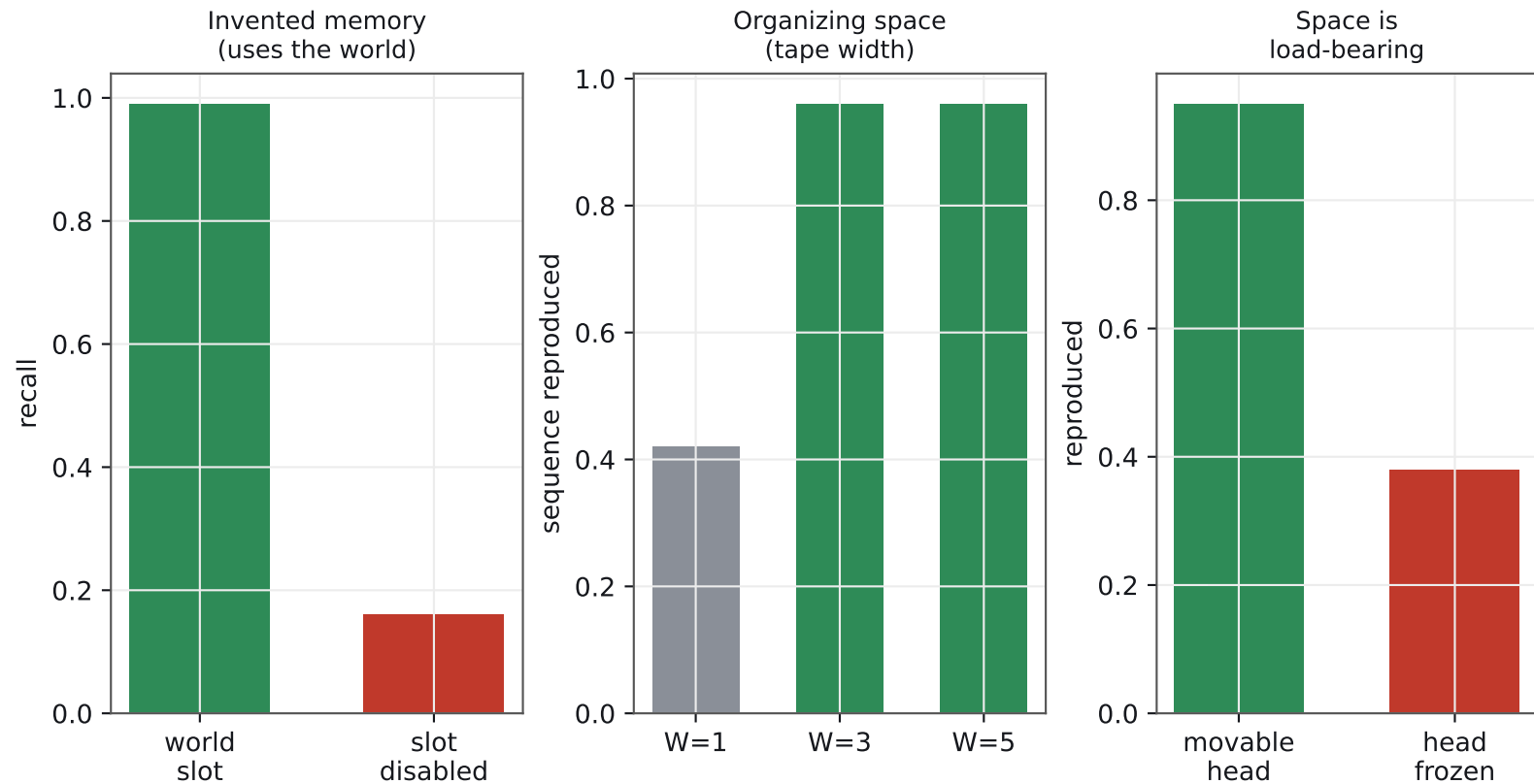
Single-word communication suffices for single concepts, so no syntax ever pays for itself. Word order becomes load-bearing only when the world has RELATIONS: events (agent, action, target) where agent and target are interchangeable, so 'A chases B' and 'B chases A' are the same symbols with opposite meaning. Survival depends on conveying who-did-what-to-whom.

Result (Figure 3)

Order-based role-marking emerges with nothing wired in (full-event accuracy ~ 0.29 vs 0.017 chance). It is genuinely USING order: scrambling the message collapses accuracy to 0.07. And the world REQUIRES order -- an order-free 'bag' listener cannot tell 'A acts B' from 'B acts A' (binding 0.19 vs the sequential channel's 0.28). Richer worlds (composite entities) push this to constituent binding.

Honest limit: absolute accuracy is modest and entities are still confused; generalisation to held-out events is partial. The MECHANISM (order emerges, is load-bearing, is required) is clean; high-accuracy grammar is not. The bottleneck became the organism: a flat genome has nowhere to HOLD a constituent -- which motivated the memory work.

Figure 4. Memory and spatial organization -- invented by a feedforward organism



4. Memory and space -- invented, not installed

The distinction that matters

The easy move is to give the organism a recurrent register and a state-update rule; then it 'has memory' and merely tunes it. The paradigm-pure question is whether the genome can BUILD a memory mechanism from primitives never designated as memory. So: a strictly FEEDFORWARD organism (no internal state), and the only persistence is in the WORLD.

Invented memory (Figure 4, left)

Given one persistent world slot and a task needing the past, the genome invents writing: it writes a cue, preserves it, and reads it back -- and it invents its OWN ENCODING (storing cue 0 as symbol 4, etc.). Ablation: 0.99 with the slot, 0.16 (chance) with it disabled. The organism is genuinely memoryless; the memory is external and self-built.

Organizing space (Figure 4, middle/right)

Given a tape and a movable head, it learns to lay a sequence ACROSS cells and scan back to reproduce it: $W=1$ caps at 0.42 (one item), $W \geq 3$ reaches 0.96. Freezing the head collapses it to the one-cell ceiling (0.95 $\rightarrow$ 0.38). It invents both the spatial layout and a per-cell cipher. This is the seed of a map -- structured knowledge held in the world.

Honest path: the spatial result required a GRADED task (reproduce a sequence; partial credit for each item). The all-or-nothing index-query version hit a reachability wall and produced no organization. The world's reward structure -- not any wired layout -- was the lever.

5. The reachability thread, and open frontiers

P1, proven repeatedly

Across syntax, memory, and spatial organization, the same pattern held: the capability the world demanded was unreachable from random genomes UNLESS partial progress was rewarded. All-or-nothing survival gave flat chance; graded survival (more of the event conveyed, more items recalled, more of the sequence reproduced = more energy) let the search climb. This is ecologically honest -- conveying or remembering more genuinely helps -- and it is not wiring in the answer; it is shaping the world so the first rung is survivable.

What is solid vs preliminary

Solid: grounding -> English; sex stabilises it; real-corpus confusion + frequency structure; order emerges and is load-bearing and required; memory is invented (ablation-proven); space is organized (ablation-proven).

Preliminary/modest: absolute syntax accuracy; compositional binding scale; memory is a low-capacity single grip (given-register case); generalisation is partial.

Open frontiers

1. CONTENT-ADDRESSED retrieval -- find by WHAT, not where -- the step that turns a tape into a memory. 2. Scaling: longer sequences, larger vocabularies, deeper relations. 3. Fusing the recurrent/world memory into the syntax worlds so binding has somewhere to live. 4. A principled measure of when a world is 'reachable' for a given capability.

Bottom line

From arbitrary signals to grounded English to word order to invented, spatially-organized memory -- each capability emerged from a world that made it the cheapest way to live, and each was the organism's own discovery. The frontier is everything that needs a richer world or a more reachable path, not a richer hand-built organism.